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BE (5th Semester)

Examination, Nov.-Dec., 2014

Branch : Mech.

MANUFACTURING SCIENCE - II

Time Allowed : Three Hours

Maximum Marks : 80

Minimum Pass Marks : 28

Note : Part (a) of each unit is compulsory. Solve any two questions from (b), (c) & (d) of each unit. Draw neat well labelled diagram wherever necessary.

UNIT - I

Q. 1. (a) What is the role of lubricant in forging ? 2

(b) What do you understand by forging process ? Give advantages of drawbacks of forging process ? 7

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(c) Describe common forging defects ? Also write the cause of each defects ? 7

(d) Explain the working of steam & air type forging press ? 7

UNIT - II

Q. 2. (a) Define angle of bite ? 2

(b) Describe wire drawing & rod drawing operation with neat sketch. 7

(c) Explain various types of rolling stand arrangements in rolling mill with neat sketch 7

(d) What is extrusion ratio ? What are its advantages & disadvantages ? 7

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UNIT - III

Q. 3. (a) Define degree of freedom ? 2

(b) Describe 3-2-1 principle of die design ? 7

(c) What are the important principle of jigs & fixtures design ? 7

(d) Explain principle of locating, clamping & indexing device ? 7

UNIT - IV

Q. 4. (a) What are the advantages of hobbing process ? 2

(b) What are the various grinding process ? Explain any one. 7

(c) Write short notes on the following (any two) : 7

(i) In feed grinding

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(ii) End feed grinding

(iii) Surface grinding

(d) Explain the various types of abrasives fo grinding wheel.

UNIT - V

Q. 5. (a) Define un-conventional machining ? 2

(b) Explain the principle of EDM process wit neat sketch ? 7

(c) What is thread rolling ? Describe proces with neat sketch. Give its advantages ? 7

(d) What are the advantages & disadvantages o unconventional machining over conventiona machining process ? 7

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